Q1

Student ID → xx ... 4 3 2 1 (last four digits)

$$A = \left[(2+1) + 1 \right] = 2$$ (fixed)

$$A = \boxed{?}$$

Write only the answer, no calculations!!!

Q2

calculated in Q1

$V_A = A + 3$ V

7.27 An NMOS transistor has $\mu_n C_{ox} = 400\ \mu A/V^2$, $W/L = 20$, $V_t = 0.5$ V, and ~~$V_A = 5$ V~~. Find g_m and r_o when (a) the bias voltage $V_{GS} = 0.75$ V, (b) the bias current $I_D = 0.5$ mA.

Q3

L = A μm

7.29 An NMOS technology has $\mu_n C_{ox}$ = 400 μA/V^2, V_t = 0.5 V, and V_A' = 6V/μm. For a transistor with ~~L = 0.5~~ μm, find r_o and the value of W that results in g_m = 2 mA/V at I_D = 0.25 mA. Also, find r_o and the required V_{GS}.

Q4

7.30 For the NMOS amplifier in Fig. P7.30, replace the transistor with its T equivalent circuit, assuming $\lambda = 0$. Derive expressions for the voltage gains v_s/v_i and v_d/v_i.

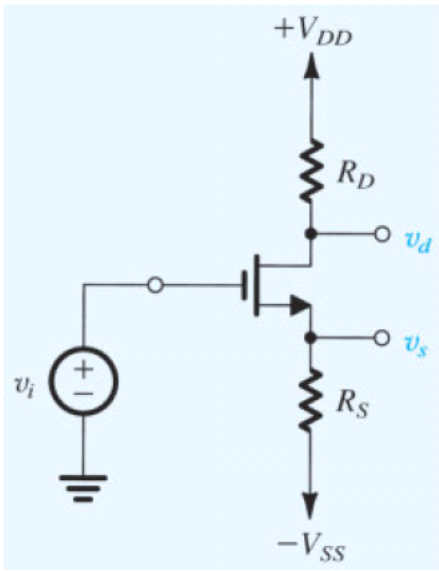


Figure P7.30

Q5

7.33 Figure P7.33 shows a discrete-circuit amplifier. The input signal v_{sig} is coupled to the gate through a very large capacitor (shown as infinite). The transistor source is connected to ground at signal frequencies via a very large capacitor (shown as infinite). The output voltage signal that develops at the drain is coupled to a load resistance via a very large capacitor (shown as infinite). All capacitors behave as short circuits for signals and as open circuits for dc.

(a) If the transistor has $V_t = 1$ V, and $k_n = 4$ mA/V^2, verify that the bias circuit establishes $V_{GS} = 1.5$ V, $I_D = 0.5$ mA, and $V_D = +7.0$ V. That is, assume these values, and verify that they are consistent with the values of the circuit components and the device parameters.

(b) Find g_m and r_o if ~~$V_A = 100$ V~~. $V_A = (5 \cdot A + 50)$ V

(c) Draw a complete small-signal equivalent circuit for the amplifier, assuming all capacitors behave as short circuits at signal frequencies.

(d) Find R_{in}, v_{gs}/v_{sig}, v_o/v_{gs}, and v_o/v_{sig}.

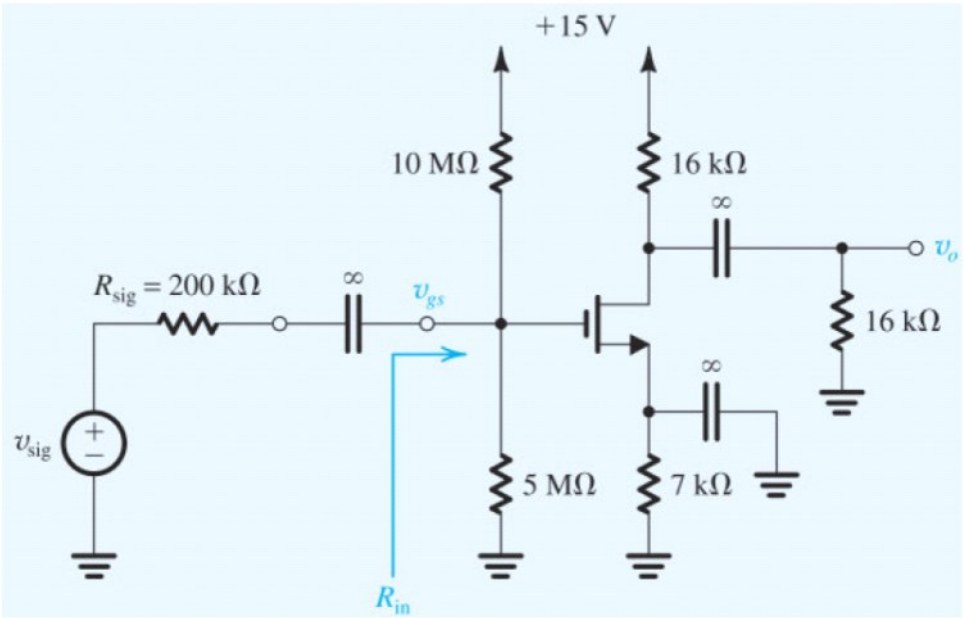


Figure P7.33

Q6

$\beta = 100 + 5$ A

7.36 A transistor with ~~$\beta = 100$~~ is biased to operate at a dc collector current of 0.4 mA. Find the values of g_m, r_π, and r_e. Repeat for a bias current of 40 μA.

Q7

$(4000 + 10^{-A})\Omega$

D 7.38 A designer wants to create a BJT amplifier with a g_m of 20 mA/V and a base input resistance of ~~4000 Ω~~ or more. What collector-bias current should she choose? What is the minimum β she can tolerate for the transistor used?

Q8

7.48 The transistor amplifier in Fig. P7.48 is biased with a current source I and has a very high β. Find the dc voltage at the collector, V_C. Also, find the value of r_e. Replace the transistor with the T model of Fig. 7.27(b) (note that the dc current source I should be replaced with an open circuit). Hence find the voltage gain v_c/v_i.

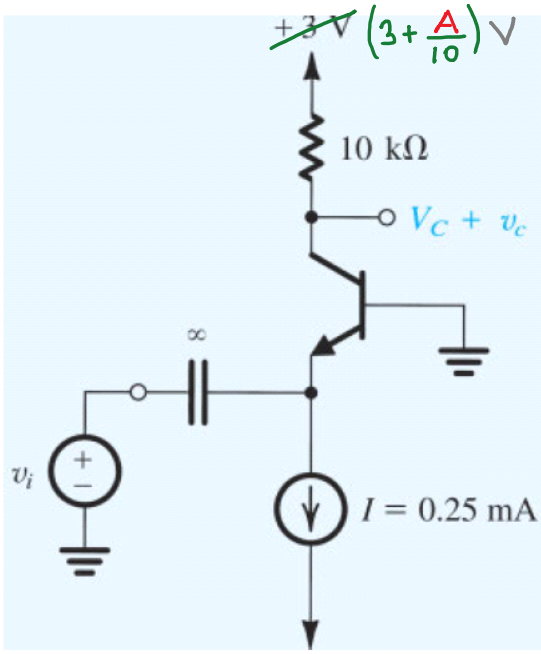


Figure P7.48

Q9

7.53 For the circuit shown in Fig. P7.53, draw a complete small-signal equivalent circuit using an appropriate T model for the BJT (use $\alpha = 0.99$). You should show the values of all components, including the model parameters. What is the input resistance R_{in}? Calculate the overall voltage gain (v_o/v_{sig}).

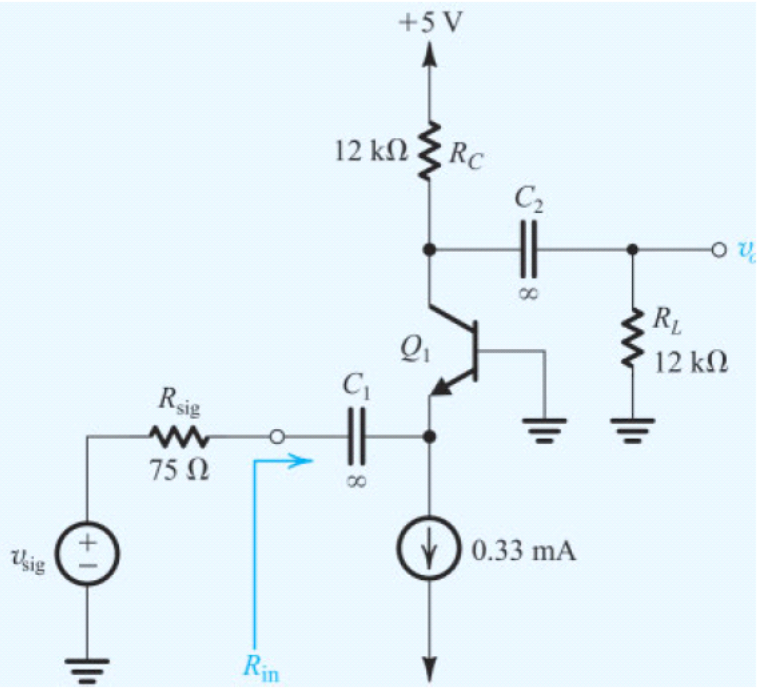


Figure P7.53

Q10

(100 + 5·A)

7.54 In the circuit shown in Fig. P7.54, the transistor has a β of ~~200~~. What is the dc voltage at the collector? Replacing the BJT with one of the hybrid-π models (neglecting r_o), draw the equivalent circuit of the amplifier. Find the input resistances R_{ib} and R_{in} and the overall voltage gain (v_o/v_{sig}). For an output signal of ±0.3 V, what values of v_{sig} and v_b are required?

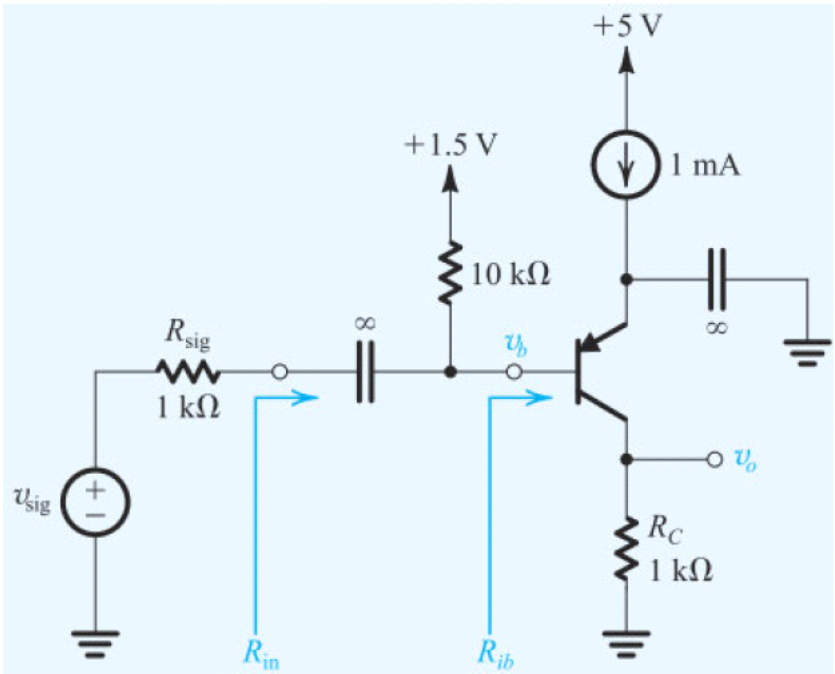


Figure P7.54